

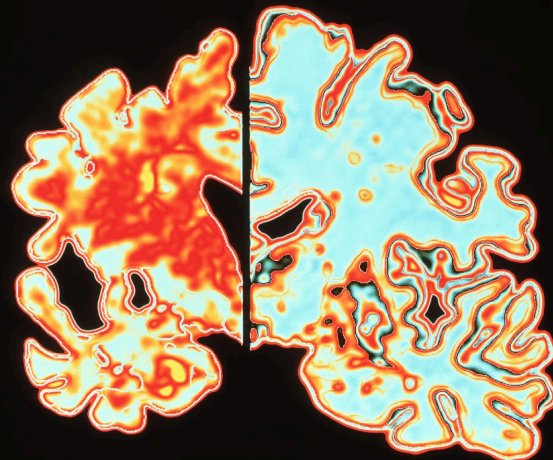
PREMIUM SCIENCE

The reason dementia rates are rising is surprisingly simple

A recent study determined the lifetime risk of developing dementia is double that of previous estimates. But there's good news here too.

By Keren Landman

Published March 19, 2025



The rate of dementia in the US will double in the years to come—at least that is what Keren Landman reports in her article, ‘The reason dementia rates are rising is surprisingly simple.’ Published on March 19, 2025, in National Geographic. Following the results established by Fang et al. (2025), the article explains how the social and medical advancements achieved by the US, brought life expectancy to raise concretely, and as a drawback dementia rates to increase overall.

As a team that includes two “Brain Science” students, we believe that dementia will be a pivotal challenge for years to come, which will be relevant especially in developed nations, where the aging population growth pose unique socioeconomic hurdles. Beyond healthcare infrastructure, the shrinking labor force and rising caregiving demands threaten broader economic stability and since the rest of our team is composed of students with background in business and economics, we are excited to face such a multifaceted issue.

Because this article focuses on the United States, governmental bodies such as the Department of Health and Human Services (HHS) serve as key stakeholders. In particular, this organization requires epidemiological data to design preventive strategies and help mitigate economic risks. Furthermore, allowing the public to view this database with the relevant research on dementia especially in the field of modifiable risk factors should reduce the economic burden of the US government in the future.

Lifetime risk and projected burden of dementia

Reference: Fang, M., Hu, J., Weiss, J., Knopman, D. S., Albert, M., Windham, B. G., Walker, K. A., Sharrett, A. R., Gottesman, R. F., Lutsey, P. L., Mosley, T., Selvin, E., & Coresh, J. (2025). Lifetime risk and projected burden of dementia. *Nature Medicine*, 31(3), 772–776. <https://doi.org/10.1038/s41591-024-03340-9>

Article Subtype: Original Research Article.

Fang et al. (2025) studies over three decades of longitudinal data (1987–2020) from participants in the Atherosclerosis Risk in Communities (ARIC) study. The population includes 15,043 participants free of dementia at the start of the study.

It provides significant quantitative results, showing that dementia lifetime risk is 42% overall, but also showing the disparities across demographic and genetic subgroups: women, non-whites and APOE ε4 are shown to be more prone to dementia.

This result is not simply caused by genetics, it proves that the probability of developing dementia is related to living conditions, especially in a country like the US, where non-hispanic whites tend to have a greater quality of life on average. This highlights the relevance of these variables, which could be potentially play a momentous role as attributes in our database.

Being a longitudinal study, it shows the observations over the course of 30 years, which can serve as a relevant tool to show the average age at which dementia occurs in our database.

In addition, the article provides the motivation for our project, given that it shows that dementia will grow as a larger problem reaching at least 1 million US citizens by 2026.

Differences in Dementia Incidence Among US Veterans.

Dintica, C. S., Bahorik, A. L., Xia, F., Boscardin, J., & Yaffe, K. (2025). Regional differences in dementia incidence among US veterans. *JAMA Neurology*, 82(8), 817–824. <https://doi.org/10.1001/jamaneurol.2025.1536>

This cohort study followed 1,268,599 dementia-free US veterans aged 65 and older within the Veterans Health Administration between 1999 and 2021 (mean follow-up 12.6 years), identifying dementia through ICD codes and using Poisson regression to compare incidence across the ten geographic regions defined by the CDC. Incidence ranged from 11.2 cases per 1000 person-years in the Mid-Atlantic to 14.0 in the Southeast; relative to the Mid-Atlantic, six regions showed rates 12% to 25% higher after demographic adjustment.

The article is relevant to our challenge in two ways. First, it justifies modelling geography as a separate entity rather than treating the United States as a single

unit: dementia incidence is demonstrably not uniform across the country, so a schema storing one national rate would conceal variation that is both real and policy-relevant. Second, the study identifies characteristics that differ across regions educational attainment, rurality, and cardiovascular comorbidity, which are candidate attributes for our geographic entity. Importantly, the regional differences persisted after adjusting for all three, meaning these factors do not fully explain the variation; that unexplained residual is itself an argument for storing them as separate queryable attributes rather than collapsing them into a single explanatory variable. Finally, the study measures incidence, which complements the prevalence and cost data in our other sources rather than duplicating it. Its main limitation is the study population: veterans, of whom only 2% were female.

Dintica, C. S., Bahorik, A. L., Xia, F., Boscardin, J., & Yaffe, K. (2025). Regional differences in dementia incidence among US veterans. *JAMA Neurology*, 82(8), 817–824. <https://doi.org/10.1001/jamaneurol.2025.1536>

Potentially Modifiable Risk Factors for Dementia and Mild Cognitive Impairment: An Umbrella Review and Meta-Analysis

Jones, A., Ali, M. U., Kenny, M., Mayhew, A., Mokashi, V., He, H., Lin, S., Yavari, E., Paik, K., Subramanian, D., Dydynsky, R., Aryal, K., Correia, R. H., Dash, D., Manis, D. R., O'Connell, M., Liu-Ambrose, T., Taler, V., McMillan, J. M., . . . Griffith, L. (2024). Potentially Modifiable Risk factors for dementia and Mild Cognitive Impairment: An Umbrella Review and Meta-Analysis. *Dementia and Geriatric Cognitive Disorders*, 53(2), 91–106. <https://doi.org/10.1159/000536643>

Article Subtype:Meta-review

Jones et al (2024) have written a meta review including 45 articles which investigated different modifiable risk factors for both mild cognitive impairment(MCI) and dementia. Before a person is diagnosed with dementia they often already have an MCI diagnosis, hence evaluating the modifiable risk factors of MCI may also help with decreasing the risk of dementia. This paper aligns with our project of building a dementia database as it fits nicely in the entity of dementia rates in which we were planning to show which factors (both modifiable and non-modifiable) increase or decrease the chance of getting dementia. This with the goal in mind of informing the public better about how they may decrease their chances of getting dementia, although it is impossible to make the chance non-existent for a single individual on the population scale the dementia rates should decline if people try to reduce their risk. This article highlights to us which modifiable factors we should include in our database. Furthermore, it shows to what extent certain factors have an influence, again something we can use to further understand this topic of modifiable risk factors.

Economic evaluations of strategies targeting pre-diagnosis dementia population

Reference: Hoang MT, Zenker AM, Saha S, Gerdtham UG, Trépel D. Economic evaluations of strategies targeting pre-diagnosis dementia populations: A systematic review. *Alzheimer's & Dementia: Translational Research & Clinical Interventions*. 2026;12(2). doi:10.1002/trc2.70236

Article Subtype: Systematic review

Hoang et al. (2026) systematically reviewed 17 economic evaluations of strategies targeting people before a dementia diagnosis — spanning identification methods, risk-reduction interventions, and other approaches such as advance care planning. The review found that most risk-reduction interventions were either cost-effective or cost-saving with specialized programmes such as that finger program (a combined diet, exercise, cognitive training, and vascular risk-control intervention) and treatments like nicotine-replacement therapy and hearing aids showing an increase in quality-adjusted life years (QALY) alongside reduced overall costs.

All this is relevant to our database because we also plan on addressing cost and value for money for therapies/treatment in the therapeutic interventions entity.